

How Do Socio-Economic Factors Influence Global Life Expectancy?



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Introduction

Why Does Life Expectancy Matter?

- Key indicator of population health
- Global population increases over time alongside improvements
- Yet large disparities remain between countries
- Socio-economic factors may explain differences



Dataset



Source

- "Life Expectancy & Socio-Economic Factors (World Bank)" dataset from Kaggle
- Variety of possible health-influencing variables



Scope

- 3,306 rows, 16 columns
- 174 countries
- 2000-2019

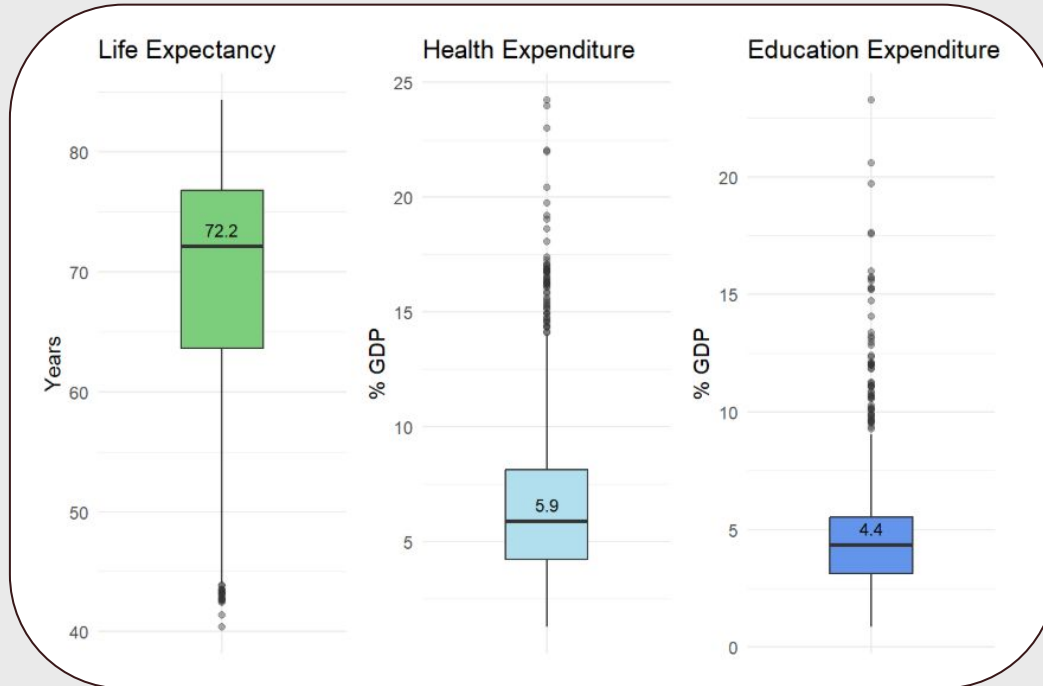


Cleaning

- Standardized column names
- Removed duplicates and empty strings
- Converted numeric strings to numeric
- Dropped rows with missing Life.expectancy

Descriptive Statistics

Distribution in the Dataset



Life expectancy varies widely across countries

Health & education spending also vary

Education expenditure has the most missing data

Research Questions

→ 1

How Does Health Expenditure Influence Life Expectancy?

→ 2

How Does Education Expenditure Influence Life Expectancy?

→ 3

How Does Life Expectancy Differ by Income Group?

Hypotheses

→

1

Health
Expenditure vs.
Life Expectancy

H_0

Health expenditure is *not* associated with life expectancy

H_1

Higher health expenditure *is* associated with higher life expectancy.

→

2

Education
Expenditure vs.
Life Expectancy

H_0

Education expenditure is *not* associated with life expectancy.

H_1

Higher education expenditure *is* associated with higher life expectancy.

→

3

Life Expectancy
by Income Group

H_0

Mean life expectancy is equal across all income groups.

H_1

At least one income group has a different mean life expectancy.

Statistical Analysis

→

1

Health
Expenditure vs.
Life Expectancy

Linear Regression

- Estimates the direction and strength of association between a continuous predictor and a continuous outcome

→

2

Education
Expenditure vs.
Life Expectancy

→

3

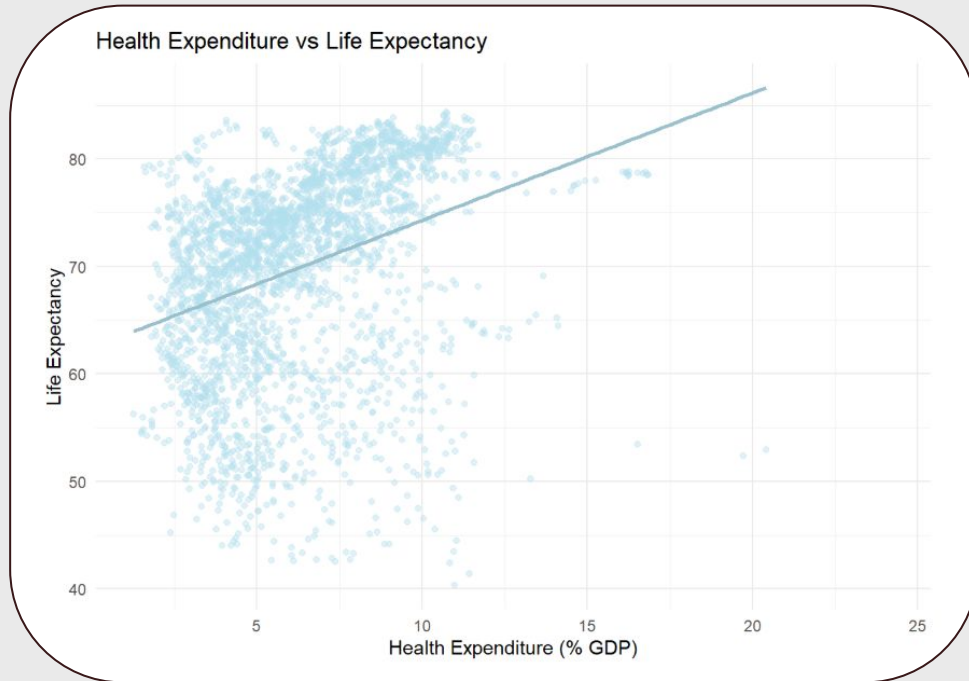
Life Expectancy
by Income Group

One-Way ANOVA

- Tests whether at least one group mean differs significantly from the others

Test Results

1. Health Expenditure Regression



Positive and statistically significant relationship

Reject null hypothesis

Health expenditure is strongly associated with life expectancy

Test Results

2. Education Expenditure Regression



Positive and statistically significant relationship

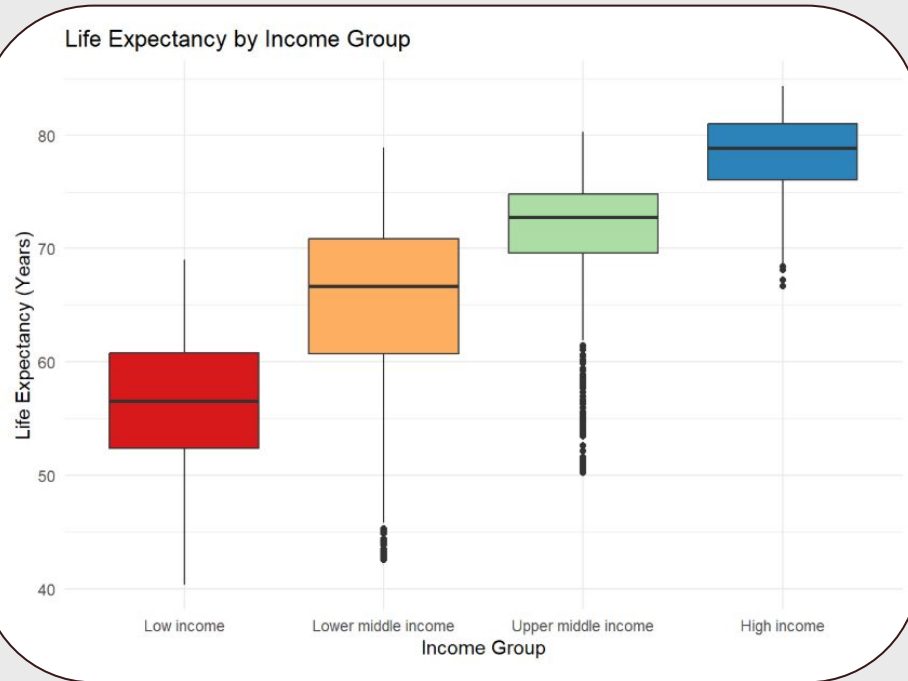
Association is weaker than for health expenditure

Reject null hypothesis

Education expenditure is strongly associated with life expectancy

Test Results

3. Income Group ANOVA



Clear increasing trend
from low- to
high-income countries

Reject null hypothesis

Mean life expectancy
differs across income
groups

Insights

1

Health

Countries that spend more on healthcare tend to have longer average lifespans

Effects are direct

2

Education

Countries that spend more on education tend to have longer average lifespans

Effects may be indirect and vary across countries

3

Income

Higher-income groups generally living longer

Reflects differences in healthcare access, living conditions, and social infrastructure

Balanced policy approach is key

Data Limitations

Not generalizable

Country-level (not individual) data

Not specific

Very simple, doesn't include other relevant socio-economic indicators

Not Representative

Missing values → reduced sample size

Not robust

Associations ≠ causation

Conclusion

Socio-economic factors strongly influence life expectancy

Health expenditure is the strongest predictor

Education and income still play key roles



→ Thank you